Final Report

Group 5

Navyesh Inaganti, Sheshinder Reddy Nandyala, Venkata Sai Polisetti,

Vinay Ravi Kumar Vandana

Department of Engineering Technology, Pittsburg State University

ETCH-810: Collaborative Projects

Dr. Derrel Fincher

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Design Report

Design Drawings and Assembly Files

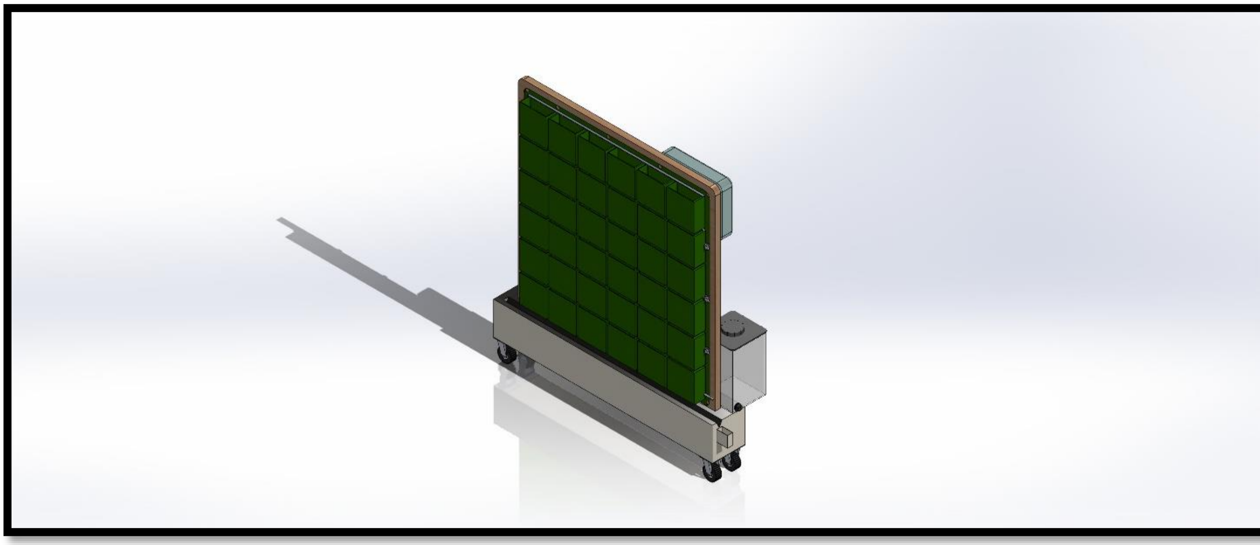
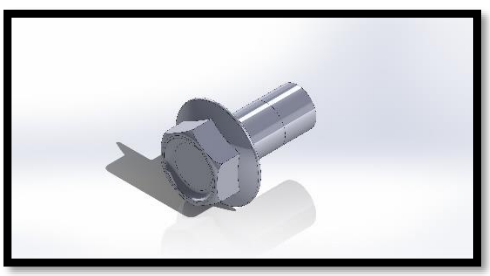


Fig 1: Green Wall CAD Assembly

Fasteners:



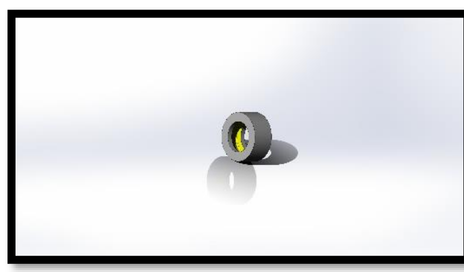


Fig 2: Fasteners

These are the fasteners used in our product, and they are 4mm diameters.

Hardware Wheels:

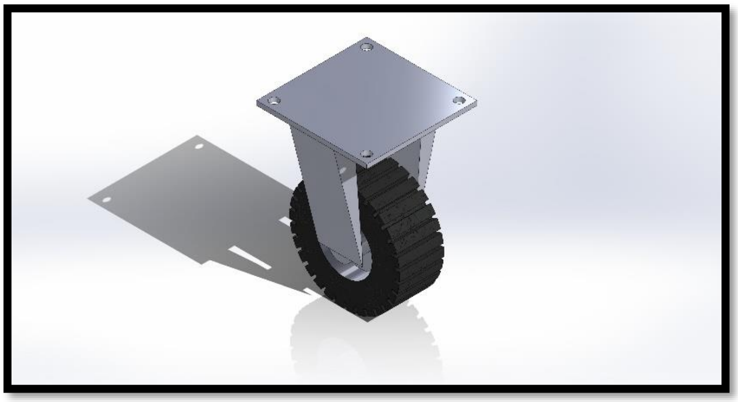


Fig 3: Hardware Wheels

The hardware wheels which we used have rubber wheels with a stainless-steel support.

Base:

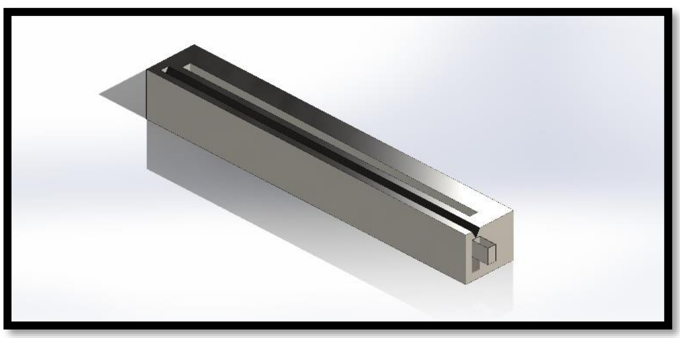


Fig 4: Base of the portable green wall

In this project we are going to use a stainless-steel Base, and the measurements are 6X12X40 (WXHXL)

Wall:

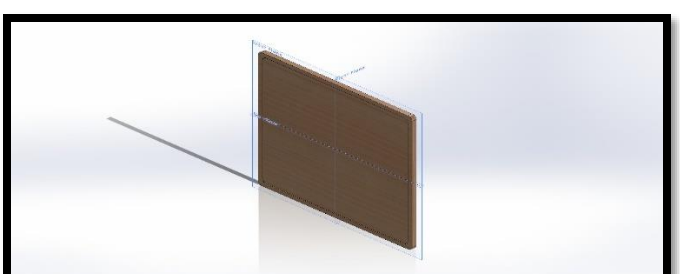


Fig 5: Cedar Wood wall frame

We are going to use Cedar wood for the wall frame and the dimensions are 36X36X1 (LXHXW) inch.

Pocket Frame:

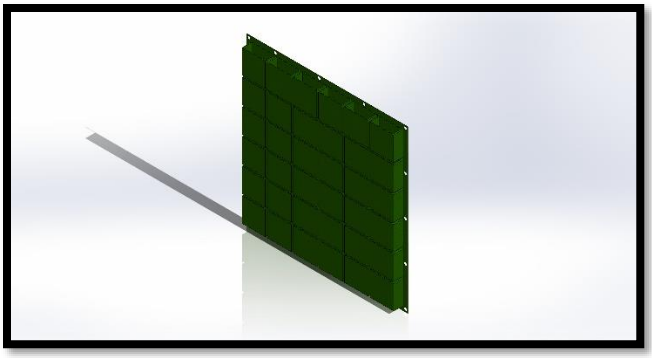


Fig 6: Fabric Pockets

In the fabric pockets we are going to place the plants, and the dimensions are 34X34 (LXH) inch.

Reservoir Tank:

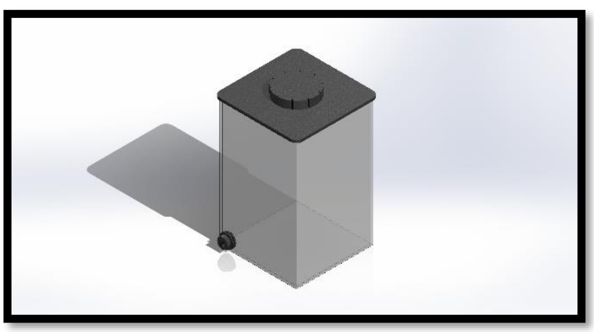


Fig 7: Reservoir Tank

The tank can hold a capacity of 1.5 gallons of water as per our requirements.

Water Dripping System:

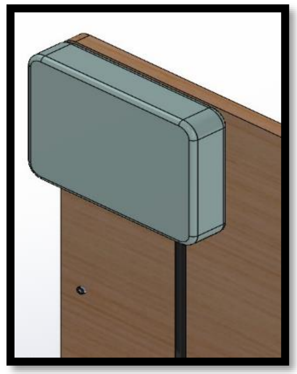


Fig 8: Water dripping System.

The water dripping system does not use any extra water and it does only use 0.04 gallons per pod/ day.

Fabrication Images:

Fig1 a, b (a on left back side of the fabricated product, b on right front side of the fabricated product)

Bill of Materials

Item Name	Description	Type	Supplier	Order Qty	Unit Type	Unit Price (USD)	Total Price (USD)	Product Link
Cedar Wood Frame	3' x 3' x 1" Cedar Frame	Raw Material	Home Depot	1	Piece	17.16	17.16	https://www.homedepot.com/p/1-in-x-4-in-x-8-ft-Square-Edge-Cedar-Boards-010408CDBOARD/319345610
Fabric Plant pods	36 Fabric Pods for plants	Component	Amazon	1	Pod	24.99	24.99	https://www.amazon.com/7Penn-Pocket-Vertical-Planter-Green/dp/B09LDZCBZK/ref=asc_df_B09LDZCBZK/?tag=hyprod-20&linkCode=df0&hvadid=693713553091&hvpos=&hvnetw=g&hv

								rand=12992719086463240173&hvpone=&hvptwo=&hvqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvlocphy=9024053&hvtargid=pla-1601077097928&mcid=e2aed045c0863e589d6ea951f27f0d5e&th=1
Water Dripping System	Automatic watering system	System	Amazon	1	Set	69.99	69.99	https://www.amazon.com/Automatic-Watering-Control-Irrigation-Shortage/dp/B0CWTWZ7T1?source=ps-sl-shoppingads-lpcontext&ref_=fplfs&smid=A1V

								UUIIFYRGGKS&th=1
Water Reservoir	1.5-gallon capacity tank	Component	Walmart	1	Piece	22.68	22.68	https://www.walmart.com/ip/Aqua-Culture-Plastic-Tank-1-5G/1747025432?wl13=79&selectedSellerId=0&wmlspartner=wlpa
Heavy-Duty Wheels	Wheels for portability	Component	Amazon	1	Set	26.89	26.89	https://www.amazon.com/Casters-Industrial-Furniture-Workbench-Hardware/dp/B09XN3P3P4/ref=asc_df_B09XN3P3P4/?tag=hyprod-20&linkCode=df0&hvadid=693270340215&hvpos

								=&hvnetw=g&hvrand=4645587739796027197&hvpone=&hvptwo=&hvqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvlocphy=9024053&hvtargid=pla-1681754331335&mcid=f0e9c5a044c83e0a814c759e0387d66e&th=1
Potting Soil	Plant soil for 36 pods	Consumable	Amazon	1	Bag	14.48	14.48	https://www.walmart.com/ip/Miracle-Gro-Potting-Mix-For-Container-Plants-Flowers-Shrubs-2-cu-ft/34621244?wmlspartner=wlpa&selectedSellerId=0

								&wl13=79&gclsrc=aw.ds&adid=222222222773462 1244_117755028 669_12420145346&wl0=&wl1=g&wl2=c&wl3=501107745824&wl4=pla-394283752452&wl5=9024053&wl6=&wl7=&wl8=&wl9=pla&wl10=8175035&wl11=local&wl12=34621244&veh=sem_LIA&gclsrc=aw.ds&gad_source=1&gclid=EAIaIQobChMI36rv5-P9iAMVqSvUAR2evjqxEAQYASABEgL_9fD_BwE

Plants	Medium sized indoor plants	Consumable	Amazon	6	Nos	4.34	26.04	https://www.amazon.com/dp/B07MVVTGHD?ref=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V&ref_=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V&social_share=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V&language=en-US&th=1
Plants	Medium sized indoor plants	Consumable	Amazon	6	Nos	5.02	30.12	https://www.amazon.com/dp/B093TP4WPB?ref=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V&ref_=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V

								&social_share=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V&language=en-US&th=1
Plants	Medium sized indoor plants	Consumable	Amazon	6	Nos	4.97	29.82	https://www.amazon.com/dp/B0B8NZYLD2?ref=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V&ref_=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V&social_share=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V&language=en-US&th=1

Plants	Medium sized indoor plants	Consumable	Amazon	6	Nos	4.81	28.86	https://www.amazon.com/dp/B0B8P1S92V?ref=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V&ref_=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V&social_share=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V&language=en-US&th=1
Plants	Medium sized indoor plants	Consumable	Amazon	6	Nos	4.95	29.7	https://www.amazon.com/dp/B0B8P1GVL5?ref=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V&ref_=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84

								V&social_share=cm_sw_r_apin_dp_DTTT7G0D3AZ3R07YV84V&language=en-US&th=1
Plants	Medium sized indoor plants	Consumable	Walmart	6	Nos	9.97	59.82	https://www.walmart.com/ip/Costa-Farms-Plants-with-Benefits-Live-Indoor-Plant-10in-Tall-Golden-Pothos-in-6in-Grower-Pot/2635061313?wmlspartner=wlpa&selectedSellerId=0&wl13=79&gclsrc=aw.ds&adid=222222222772635061313_117755028669_12420145346&wl0=&wl1=g&wl2=c&

								wl3=501107745824&wl4=pla-394283752452&wl5=9024053&wl6=&wl7=&wl8=&wl9=pla&wl10=8175035&wl11=local&wl12=2635061313&veh=sem_LIA&gclsrc=aw.ds&gad_source=1&gclid=EAIaIQobChMI8PGu9ub9iAMVbW5_AB2dFAw1EAQYASABEgIRIPD_BwE
Pump	Low power pump (15W/day)	Component	Amazon	1	Nos	35.44	35.44	https://www.amazon.com/FIERRG-Adjustable-Submersible-Fountain-Accessories/dp/B0BYMBNPXZ

Fasteners & Joints	Screws, brackets	Component	Walmart	3	Pack	1.24	3.72	https://www.walmart.com/ip/Hillman-Wood-Screws-8-x-1-Flat-Phillips-Zinc-Plated-Steel-Pack-of-22/478895172?wmlspartner=wlpa&selectedSellerId=0&wl13=72&gclsrc=aw.ds&adid=22222222277478895172_117755028669_12420145346&wl0=&wl1=g&wl2=c&wl3=501107745824&wl4=pla-394283752452&wl5=9024053&wl6=&wl7=&wl8=&wl9=pla&wl10=8175035&wl11

								=local&wl12=478895172&veh=sem_LIA&gclsrc=aw.ds&gad_source=1&gclid=EAIaIQobChMI5sabpuT9iAMVfjbUAR0oEA03EAQYAyABEgJDN_D_BwE

Design Overview for The Portable Green Wall

Purpose and Features

The main objective of the design is to develop a green wall system that is maintenance free and friendly to the environment and designed to be installed in indoor environments the system meets the current requirements which include stress relief air quality, and the overall appearance of the space key features include.

- **Automatic water dripping system:** Accurately provides 0.04 gallons of water daily for each pod and therefore is very efficient.
- **Water recycling system:** Excess water is collected it can be used again thus the water use efficiency is at 95%.
- **Portability:** It has wheels that are designed to carry heavy loads which makes it suitable for flexible positioning within indoor environments.

- **Durability:** Constructed of cedar wood which is very durable and has a long life of 25 years and above with natural resistance.
- **Compact design:** It has the size of three feet by 3 feet by 1 thus suitable for spaces which have limited space.

Material and Structural Design

- **Frame:** We chose cedar wood because it is strong durable and eco-friendly. The measurement 3 by 3 feet by 1 foot ensures that the structure is strong enough to support the components.
- **Plant pods:** There are 36 fabric plant pods that can accommodate 0.05 cubic feet of soil enough for medium sized plants.
- **Reservoir:** The water is provided by a 1.5-gallon tank which is connected to the automatic dripping system.
- **Heavy duty wheels:** Provide stability and the mobility of the system placement adjustments have enhanced the stability in the mobility of the system and have also made it easier to move around.

Functional highlights

- **Ease of installation:** It is very easy to install this house indoors and the setup is rather simple.
- **Esthetic and environmental benefits:** It improves the quality of air, helps in stress reduction and provides greenery to the indoor environment.
- **Low maintenance:** Some of the features that reduce the need for manual intervention include self-watering and water recycling.

Design evolution

When going from the prototype stage to the production stage:

- The number of plant pods was also cut down from 36 to 30 to improve the water drainage and recovery.

- The location of the wheels as well as the position of the reservoir was also enhanced with a view to enhancing stability.
- The materials used were also enhanced to improve the appearance and the durability of the product.

Sustainability and quality assurance

Sustainability is one of the key principles of resources which have reduced waste followed:

- Water recycling reduces waste and environmental impact.
- Durable cedar would ensure a long-life cycle.
- Minimal resource usage supports eco-friendly objectives.

The following are quality assurance measures:

- Checking the performance of the water dripping system and the recycling mechanisms on a regular basis.
- Structural soundness and looks of the structure are also checked from time to time.

User benefits

- **Health and Wellness:** Helps in enhancing the quality of air and acts as a stress Buster.
- **Minimal maintenance:** Automated processes decrease the time and energy spent by the users.
- **Flexibility:** It is flexible and can be moved around and can be used in different rooms in the house.

Assembly Of Portable Green Wall with Automatic Water Dripping System

1. **Frame construction**
    - **Material:** The main structural members are made of cedar wood which is durable and does not easily decay.

- **Dimensions:** The wooden frame was made to measure to the size of three feet high 3 feet wide and one inch thick.
- **Assembly process**
  1) The cedar wood pieces are first made into shapes that are in line with the desired sizes.
  2) In this structure fasteners and brackets are used in the joints to increase the strength of the structure.
  3) The frame is then given a complete aesthetic makeover with paint that is waterproof to ensure that it does not get ruined by moisture.

2. Plant pods installation.

- **Components:** 36 fabric plant pods with a capacity to hold 0.05 cubic feet of soil.
- **Placement:**
  1) The fabric pods are placed on the wooden frame in a manner that is equally spaced all through with the aim of having a good design and structure.
  2) The pods are fixed firmly with stainless steel screws to prevent the pods from detaching when filled with soil and plants.

3. **Water dripping system installation.**

- **System description:**
  1) The automatic water dripping mechanism which provides 0.04 gallons of water per pod on daily basis.
  2) A 1.5-gallon water tank is used to store water in the system.
- **Installation steps:**
  1) The water reservoir is placed at the bottom of the structure to ensure that the center of gravity is low.
  2) Drip irrigation pipes are passed through the frame and then connected to each pod.

3) The system is checked to see if it is delivering the right amount of water to each plant part in a leak free manner.

4. **Water recycling and overflow system**

- **Mechanism:** Collects excess water and returns it back to the reservoir for the next use.
- **Assembly process**

1) A collection tray is provided at the bottom of the pods to ensure that water does not accumulate on the surface.
2) A short flexible tube leads from the tray to the reservoir thus creating a closed loop water recycling system.

5. **Heavy Duty Wheels Installation**

- **Purpose:** to increase the mobility of the structure so that it can be easily moved from one place to another.
- **Installation steps:**

1) In this case 4 heavy duty wheels are incorporated at the bottom of the with the help of steel brackets.
2) The wheels also have locks that help to protect the green wall from shaking when placed in a fixed position.
3) Placement adjustments help in ensuring that the weight is distributed equally thus enabling easy movement.

6. **Soil and plant set up.**

- **Preparation:**

1) Each pod contains rich potting soil for supporting medium sized indoor plants.
2) The plants are chosen in a way that they can comfortably grow in the indoor environment and meet the specific needs of the user.

- **Process:**
  1) The soil is placed into the pods in a shaker-like manner to support the equal growth of plants.
  2) The plants are placed in the pods in such a way that they are held firmly in place to have the best view and function.

7. **Final adjustments and testing**

- **System calibration:**
  1) The automatic water dripping system is turned on and the flow rates of water are checked to ensure that it is constant.
  2) The water recycling system is also checked to make sure that any water that is beyond the required is found as well collected in the reservoir.
- **Aesthetic enhancements:**
  1) The structure of the plants is assessed to determine if the design is aesthetically pleasing overall.
- **Quality assurance:**
  1) The last stress tests are performed to check the structural integrity of the structure and its performance.

8. **Packaging for delivery**

- **Disassembly:**
  1) The big parts are unfolded for easy transportation and to ensure that they are not damaged during transportation.
- **Packaging:**
  1) Each part is items packaged to ensure that it is not damaged during transportation.

Product Specifications for The Portable Green Wall

1. **Dimensions and frame**

   - **Overall dimensions:** It measures 3 feet tall, 3 feet wide and one inch thick only.
   - **Frame Material:** The frame is made of cedar wood, and the following reasons have been taken.
     1) **Durability:** It has an average lifespan of 25 years and is not easily damaged by decay or termites.
     2) **Eco friendliness:** A renewable resource which is consistent with the objectives of sustainable design.
   - **Weight:** It is estimated to weigh about 140 lbs. But is strong enough to support itself yet easy to move around.

2. **Plant pods**

   - **Quantity:** W have used 36 fabric plant pods out of which I changed the number to 30 in the final design to make the water usage effective.
   - **Material:** Stainless steel reinforced felt fabric to make it durable and able to be flexed.
   - **Capacity:** It can accommodate 0.05 cubic feet of soil which is enough for small to medium sized indoor plants.
   - **Arrangement:** Spaced equally in the frame to ensure that the structure is symmetrical and well balanced.

3. **Automatic water dripping system.**

   - **Water reservoir:**
     1) Capacity it has a capacity of 1.5 gallons of water, enough for several days of use.
     2) Placement the base is mounted at the bottom of the structure to decrease the overall balance of the structure.

- **Drip rate:**
  1) Each pot is supplied with 0.04 gallons of water daily to ensure that water is used properly and in the right amounts.
- **Efficiency:**
  1) This system is efficient in the use of water as it is 95% effective in the use of water.
- **Automation:**
  1) The entire system is automatic and comes with preprogrammed schedules that do not require much human input.

4. Water recycling system

- **Mechanism:**
  1) It comes with the feature of collecting water from the plant pods and then returning the same water back to the reservoir.
- **Impact**
  1) The use of the system in promoting sustainability is that it helps to prevent water wastage.
  2) Enables the sustained operation of the system through the closed loop concept of water use.

5. Portability

- Heavy duty wheels:
  1) Number of wheels for the base there are four wheels.
  2) Features these include locking mechanisms to ensure that the structure is not moved around aimlessly when it is placed in each position.
  3) Placement the placement of the wheels is in a way that the weight is distributed equally thus enabling easy movement.

6. **Maintenance**

- **Self-watering:**

  1) The automated system reduces the need for daily watering and thus the effort that is required from the design users.

- **Ease of Cleaning:**

  1) It is easy to disassemble and take apart the various components for cleaning and for maintenance.

7. **Sustainability**

- **Material selection:**

  1) The use of sustainable materials such as cedar wood and fabric plant pods that are renewable and long lasting.

- **Water conservation:**

  1) This system of automatic water dripping and recycling helps to reduce the negative effects on the environment.

8. **Functional features**

- **Air quality improvement:**

  1) Contains various types of plants which can purify the indoor air.

- **Stress relief:**

  1) The integration of plants into the indoor environment has been known to have positive effects on mental health and increase productivity at work.

- **Ease of use:**

  1) A very user-friendly design that can be used by both beginners and experienced gardeners.

9. **Aesthetic appeal**

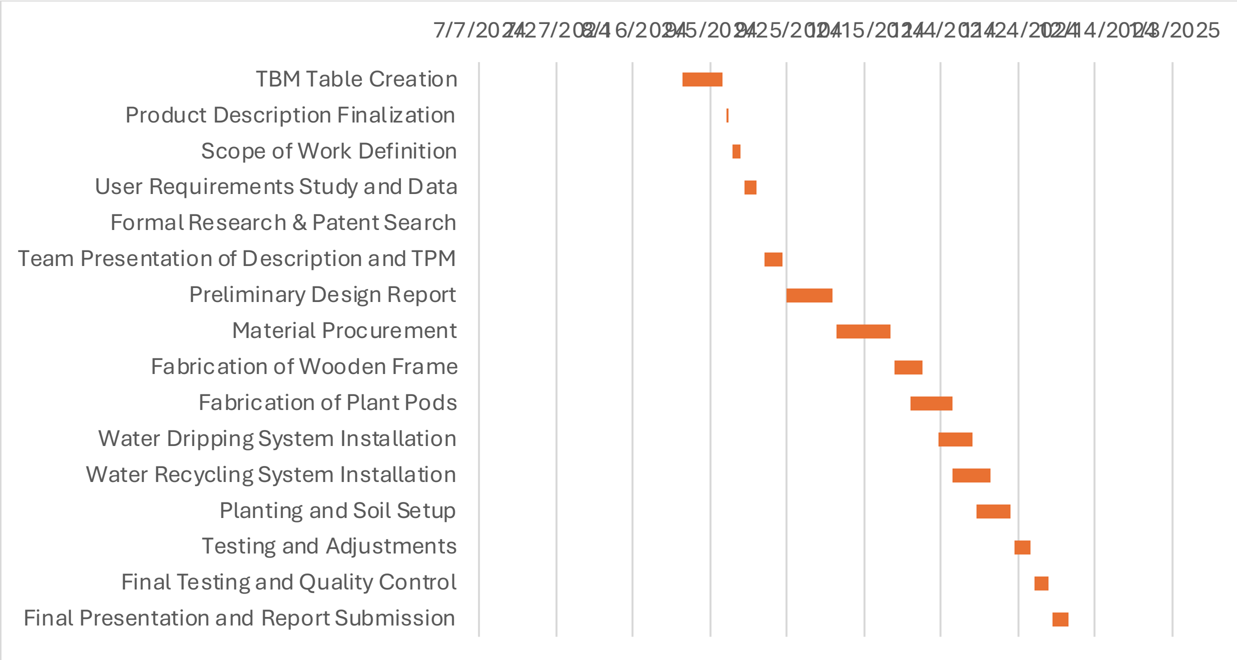
- **Design:**
    1) It is sleek and modern and would be appropriate for use in offices, homes and other indoor areas.
- **Customizable plant arrangements:**
    1) This enables the users to arrange the green wall in any way they want.

Schedule (Gantt Chart):

S. No.	Milestone	Start Date	End Date	Number of Days	Cumulative Days
1.	**TBM Table Creation**	8/29/2024	9/8/2024	10	10
2.	**Product Description Finalization**	9/9/2024	9/10/2024	1	11
3.	**Scope of Work Definition**	9/11/2024	9/13/2024	2	13
4.	**User Requirements Study and Data**	9/14/2024	9/17/2024	3	16

5.	**Formal Research & Patent Search**	9/18/2024	9/18/2024	0	16
6.	**Team Presentation of Description and TPM**	9/19/2024	9/24/2024	5	21
7.	**Preliminary Design Report**	9/25/2024	10/7/2024	12	33
8.	**Material Procurement**	10/8/2024	10/22/2024	14	47
9.	**Fabrication of Wooden Frame**	10/23/2024	10/30/2024	7	54
10.	**Fabrication of Plant Pods**	10/27/2024	11/7/2024	11	65
11.	**Water Dripping**	11/3/2024	11/12/2024	9	74

	System Installation				
12.	**Water Recycling System Installation**	11/7/2024	11/17/2024	10	84
13.	**Planting and Soil Setup**	11/13/2024	11/22/2024	9	93
14.	**Testing and Adjustments**	11/23/2024	11/27/2024	4	97
15.	**Final Testing and Quality Control**	11/28/2024	12/2/2024	4	101
16.	**Final Presentation and Report Submission**	12/3/2024	12/7/2024	4	105



Manufacturing Plans for The Portable Green Wall

1. Material procurement

The proper acquisition of quality materials is one of the most important initial strategies.

- **Wooden frame:** cedar wood three by three feet by 1 has been used as it is durable, has a nice finish and is eco-friendly.
  1) **Supplier:** Local lumber stores or online specialty wood retailers.
  2) **Lead time:** It will take about two weeks.
- **Plant pods:** The pods are made of stainless steel felt fabric 36 pods at the beginning and 30 pods in the final design.
  1) **Supplier:** Gardening suppliers are online retailers.
  2) **Lead time:** 2 weeks.
- **Water system components:**
  1) Automatic water dripping system (pipes, connectors, control mechanisms).
  2) Water reservoir tank (1.5-gallon capacity).

3) **Supplier:** the suppliers of irrigation and plumbing equipment.

4) **Lead time:** 2 weeks.

- **Wheels:** These include wheels with high load bearing capacity and locking mechanisms to make them steady

1) Supplier hardware suppliers.

2) Lead time: 2 weeks.

2. **Frame Fabrication**

- **Wood cutting and assembly:**

1) Cut cedar wood to the right size 3 long 3 feet wide and one inch thick.

2) The construction of the frame is that it has stainless steel fasteners and brackets that hold it.

3) The surface of the table is painted with waterproof paint for better appearance and durability.

- **Testing:**

1) Make sure that the frame has the capacity of holding the weight of the plant pods water reservoir and the watering system.

3. **The fabrication of plant pods**

- **Installation:**

1) The fabric plant pods are then fixed onto the wooden frame with the help of brackets that have been strengthened.

2) Place the pods in a regular pattern to make sure that the weight of this structure is distributed evenly and to allow for equal water distribution.

- **Quality check:**

1) Make sure that the fabric used is strong and well-fixed to the frame.

4. **Water dripping system installation.**

- **Set up:**

1) Connect the 1.5-gallon water reservoir to the automatic dripping system.
2) Laying pipes and connectors all from all pods.
3) For water to recycling provide mechanism water has evenly been to put in place to capture any water that may be wasted.

- **Testing:**

1) To check the flow test to confirm that water flow is constant at 0.04 gallons in a day to each pod.
2) To check the efficacy of the recycling system.

5. **Water recycling and overflow system**

- **System installation:**

1) Place a tray at the bottom of the frame for water accumulation.
2) Promote the reuse of water. The tray is connected to the reservoir using tubes that form a closed loop system.

- **Verification:**

1) Test for leaks and ensure effective water redirection.

6. **Wheel attachment for portability**

- **Assembly:**

1) The base of the frame should be fitted with four heavy duty wheels with the help of brackets.
2) The placement of the wheels should be done in a way that ensures that the weight of the whole structure is well distributed.
3) Check the mobility of the wheels and the locking mechanism.

7. **Final Assembly and Adjustments**

- **Integration:**

  1) Integrate the wooden structure the plant cradles the waterworks and the wheels in one structure.

- **Testing:**

  1) The entire system must be tested to check if it is working properly, including water flow recycling and structural integrity.

- **Aesthetic checks:**

  1) Place the plants in a manner that would be aesthetically pleasing and imbalance.

8. **Quality assurance and Packaging**

- **Inspection:**

  1) Perform final inspections for durability operational consistency find authentic standards.

- **Packaging:**

  1) Disassemble larger components for easier transport.
  2) Use protective materials to prevent damage during shipping.

9. **Manufacturing timeline**

- **Gantt chart overview:**

  1) Material procurement: 2 weeks October 8th to October 22nd, 2024.
  2) Frame construction: 1 week October 23rd to October 30th, 2024.
  3) Plant pod fabrication: 11 days October 27th to November 7th, 2024.
  4) Water system installation: 9 days November 3rd to November 12th, 2024.
  5) Testing and adjustments: 1 week November 23rd to November 27th, 2024.

10. **Scalability for Production**

- **Batch size:** Initially scaled for small batches to validate manufacturing processes.
- **Automation:** Explore semi-automated fabrication for larger scale production.
- **Supplier contracts:** Establish long-term agreements with reliable suppliers for consistent quality.

Quality Assurance Plan

1. **Quality assurance overview**

   The QA process is to make sure that each component of the portable green wall is as efficient as it needs the main objectives are:

   - To maintain consistency and performance and aesthetics.
   - To minimize field failures and customer complaints.
   - To uphold the sustainable and user-friendly ethos of the design.

2. **QA during material procurement**

   - **Inspection Criteria:**

     1) **Cedar wood:** Criteria must obey cedar free from warping cracking and defects to provide strength and durability.
     2) **Fabric plant pods:** Check for any tears in the fabric or if it is evenly made to ensure efficiency that of it that is water not tank developed in who's terms of which it can its lead capacity to the leakage.
     3) **Water reservoir:** Strength of the water material reservoir used to check the ability of the tank not to develop leaks.
     4) **Dripping system components:** Make sure that all the connectors' pipes and valves used in the system meet the set standards.
     5) **Wheels:** Check the flexibility of the wheels and the ability of the wheels to roll smoothly and freely with or without load as well as the effectiveness of the locks.

- **Testing frequency:**
    1) Sample inspection means the 10% of the material receipt should be checked to determine whether there are defects or not.
    2) Certification makes sure that the suppliers can produce quality certificates for the major parts of the product.

3. **QA during manufacturing**

- **Wooden frame construction:**
    1) Check the dimensions of the cut parts to make sure that they meet the required tolerance of ±1 mm.
    2) Check the joints and fasteners for the structural integrity of the structure for the subjected load conditions.
- **Plant pod installation:**
    1) Make sure that the pods are fixed properly and in a proper manner on the frame.
    2) Verify pod capacity to hold soil and plants without sagging.
- **Water dripping system:**
    1) End pressure and flow tests should be carried out to confirm that water supply to each pod has between 0.04 and 0.05 gallons per day.
    2) Check for any losses in the piping as well as the connections to the reservoir.
- **Water recycling system:**
    1) Make sure that the leftover water is being captured and then discharged back into the reservoir without any loss.
    2) Check the position of the tray and the portability tubing testing.

4. **QA for Final Assembly**

- **Portability Testing:**

1) Check on the condition of the wheels and the locking mechanisms.
2) It should be noted that when the wall is in a stationary position it should be in a stable condition and in motion.

- **Durability tests:**
  1) Subject the structure to a series of tests whereby the frame is subjected to a weight that is equivalent to that of filled pods in a full reservoir to check on the stability of the frame.
  2) Make sure that the structure is not weakened and will not break even when used frequently.
- **Aesthetic inspection:**
  1) Assess the final look to ensure that there is consistency in the placement of plants as well as the quality of finishing.

5. Testing protocols

- **Water distribution testing:**
  1) Use sensors or manual observation to confirm even water distribution across all pods.
  2) Operate it for a continuous 48 hours to check the performance to see if there are any faults.
- **Water recycling efficiency:**
  1) To quantify the quantity of water reused in the given time to check on the efficiency of the system to recycle 95% of water.
- **Stress and load testing:**
  1) Imitate the maximum load conditions to test the products capability to meet the expected usage conditions.
- **Environmental testing:**

1) Subject the green wall in different temperature and humidity conditions to check its suitability for use in indoor environments.

6. **QA for production scalability:**
   - **Batch testing frequency:**
     1) Initially test every 10th unit during the initial production phase.
     2) Taper down to one for every 50 units after the QA process is standardized.
   - **Lot testing:**
     1) Choose random components supplied by the suppliers EG every 100 units of plant pods or wood panels to subject them through a series of tests.
   - **Customer feedback loop:**
     1) Collect information from the first users the track and fix the problems that may appear.
7. **Documentation and reporting**
   - **QA checklists:**
     1) Ensure that you keep detailed records of the assessments made including the dimensions water flow rates and the low-test results.
   - **Defect tracking:**
     1) The following is a defect tracking system that is used to identify the trends and then come up with the necessary solutions.
   - **Final approval:**
     1) A QA engineer approves the product only after all the tests have been conducted and have produced the desired results.
8. **Ongoing quality assurance**
   - It is therefore important to have a field failure analysis program to assess the performance of the product in the field and the satisfaction of the users.

- It also puts in place a return and repair mechanism in case of defects found after delivery has been made.
- A periodic assessment of the QA processes and the testing methodologies should be made to capture any improvements.

Fig 2: Team Avengers with Final product.